



THE BASICS OF ACCELEROMETER CALIBRATION

THE MODAL SHOP

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THE BASICS OF ACCELEROMETER CALIBRATION



Learning Objectives:

- What is Calibration
- Why Calibrate
- Types of Calibration
- What is Uncertainty
- What is a Field Check

CALIBRATION OVERVIEW

“Property of a measurement result whereby the result can be related to a reference (typically the SI) through a documented unbroken chain of calibrations, each contributing to measurement uncertainty.”

~NCSL International

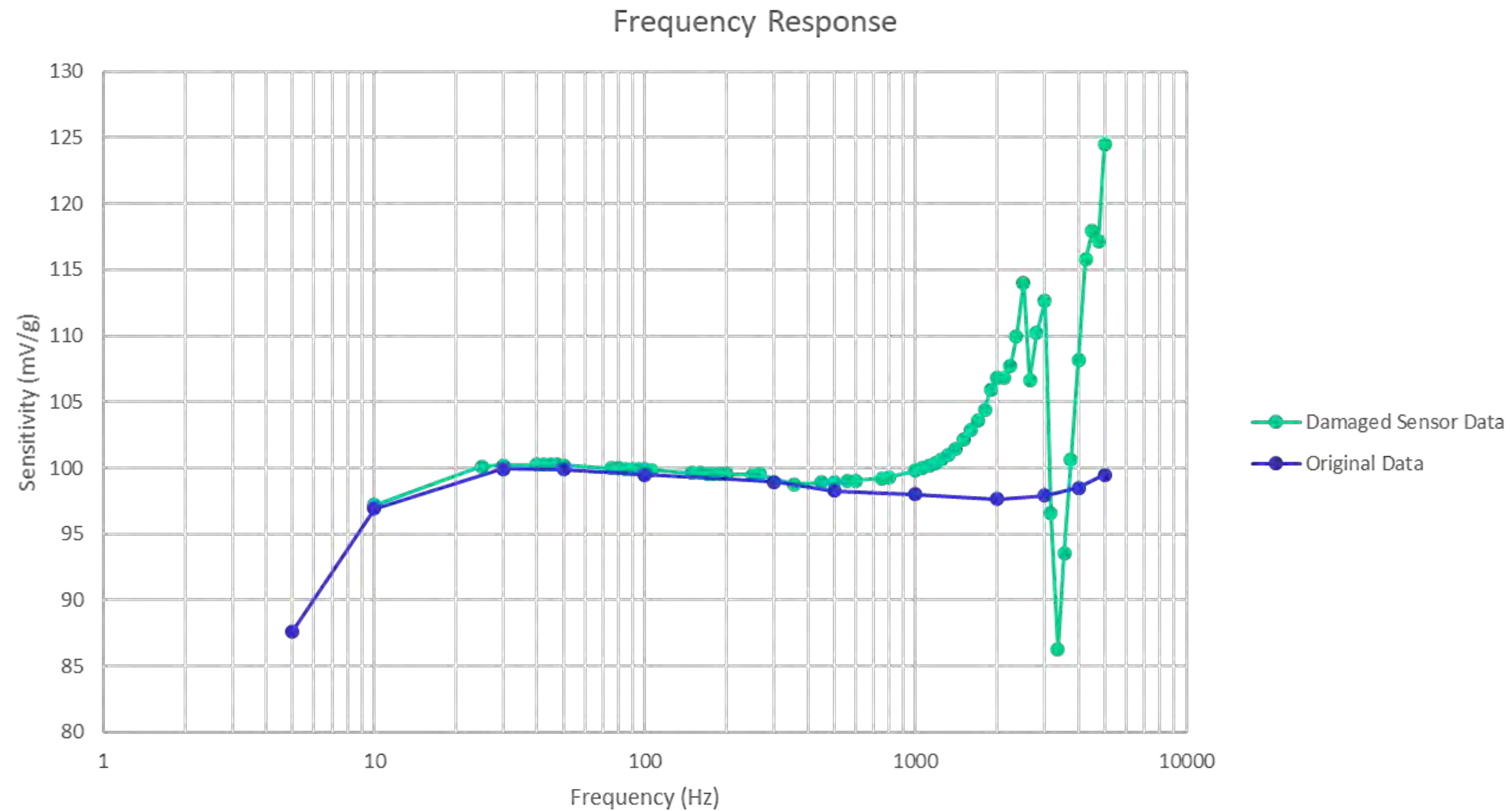
CALIBRATION OVERVIEW

- Why Calibrate?
 - To ensure the sensor performs according to the specification and to provide meaning to the electrical output
- Why Recalibrate?
 - Good measurement practice that helps to ensure accurate results and that the sensor is still “healthy”
 - Reference sensitivity, frequency response, output bias level and resonant frequency



- Often required by ISO 9001, QS9000 or contractual reasons
- Risk Mitigation (17025:2017)

WHY RECALIBRATE?



WHAT IS ACCELEROMETER CALIBRATION?

- Typical Manufacturer Calibration Includes:
 - Reference sensitivity
 - Frequency response
 - Resonant frequency
 - Linearity
 - Transverse sensitivity
 - Output bias or ZMO voltage
 - Time constant

Model Number	ICP® ACCELEROMETER		
353B04			
Performance	ENGLISH	SI	
Sensitivity(± 5 %)	10 mV/g	1.02 mV/(m/s²)	[2]
Measurement Range	± 500 g pk	± 4905 m/s² pk	
Frequency Range(± 5 %)	1 to 7000 Hz	1 to 7000 Hz	
Frequency Range(± 10 %)	0.7 to 11,000 Hz	0.7 to 11,000 Hz	
Frequency Range(± 3 dB)	0.35 to 20,000 Hz	0.35 to 20,000 Hz	
Resonant Frequency	≥ 38 kHz	≥ 38 kHz	
Broadband Resolution(1 to 10,000 Hz)	0.003 g rms	0.03 m/s² rms	[1]
Non-Linearity	≤ 1 %	≤ 1 %	[3]
Transverse Sensitivity	≤ 5 %	≤ 5 %	[4]
Environmental			
Overload Limit(Shock)	± 10,000 g pk	± 98,100 m/s² pk	
Temperature Range(Operating)	-65 to +250 °F	-54 to +121 °C	
Temperature Response	See Graph	See Graph	[1]
Base Strain Sensitivity	≤ 0.0005 g/με	≤ 0.005 (m/s²)/με	[1]
Electrical			
Excitation Voltage	18 to 30 VDC	18 to 30 VDC	
Constant Current Excitation	2 to 20 mA	2 to 20 mA	
Output Impedance	≤ 100 ohm	≤ 100 ohm	
Output Bias Voltage	8 to 12 VDC	8 to 12 VDC	
Discharge Time Constant	0.5 to 2.6 sec	0.5 to 2.6 sec	
Settling Time(within 10% of bias)	<5 sec	<5 sec	
Spectral Noise(1 Hz)	2800 μg/√Hz	27,468 (μm/sec²)/√Hz	[1]
Spectral Noise(10 Hz)	700 μg/√Hz	6867 (μm/sec²)/√Hz	[1]
Spectral Noise(100 Hz)	180 μg/√Hz	1766 (μm/sec²)/√Hz	[1]
Spectral Noise(1 kHz)	64 μg/√Hz	628 (μm/sec²)/√Hz	[1]
Physical			
Sensing Element	Quartz	Quartz	
Sensing Geometry	Shear	Shear	
Housing Material	Titanium	Titanium	
Sealing	Welded Hermetic	Welded Hermetic	
Size (Hex x Height)	0.50 in x 1.14 in	12.7 mm x 29.0 mm	
Weight	0.38 oz	10.5 gm	[1]
Electrical Connector	10-32 Coaxial Jack	10-32 Coaxial Jack	
Electrical Connection Position	Top	Top	
Mounting Thread	10-32 Female	10-32 Female	

Optional versions
e)

B - Low bias elec
Output Bias Voltage
Excitation Voltage
Constant Current
Measurement Range

J - Ground Isolate
Frequency Range
Frequency Range
Resonant Frequency
Electrical Isolation
Size - Hex x Height

Q - Extended disc
Frequency Range
Frequency Range
Discharge Time Constant
Settling Time(within
Supplied Accessory
cal from 0.5 to 10)

W - Water Resistant
Electrical Connector
Electrical Connector

NOTES:
[1] Typical
[2] B and Q options
[3] Zero-based, lead
[4] Transverse sensitivity
[5] See PCB Declaration

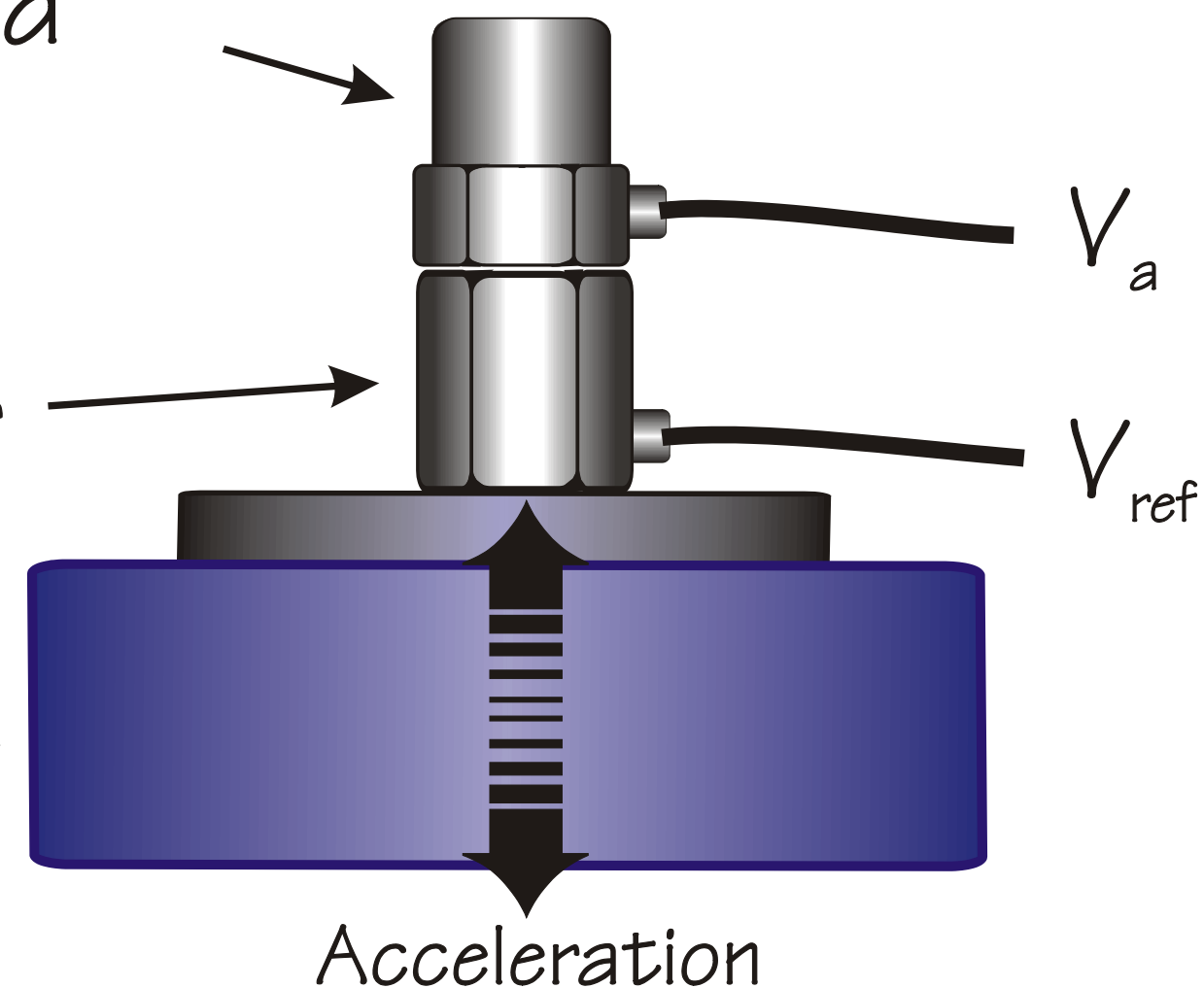
CALIBRATION METHODS: REFERENCE SENSITIVITY

- Test sensor mounts “back-to-back” with a reference accelerometer that has been previously calibrated by primary means or by a recognized laboratory
- Reference Sensitivity is determined at reference frequency (100 Hz or 159.2 Hz typical)

Accelerometer
to be calibrated

Calibration
Standard
Accelerometer
(Known sensor S_{ref})

Shaker



$$A_a = A_{ref}$$

$$\frac{V_a}{S_a} = \frac{V_{ref}}{S_{ref}}$$

$$S_a = S_{ref} \frac{V_a}{V_{ref}}$$



CALIBRATION METHODS: FREQUENCY RESPONSE

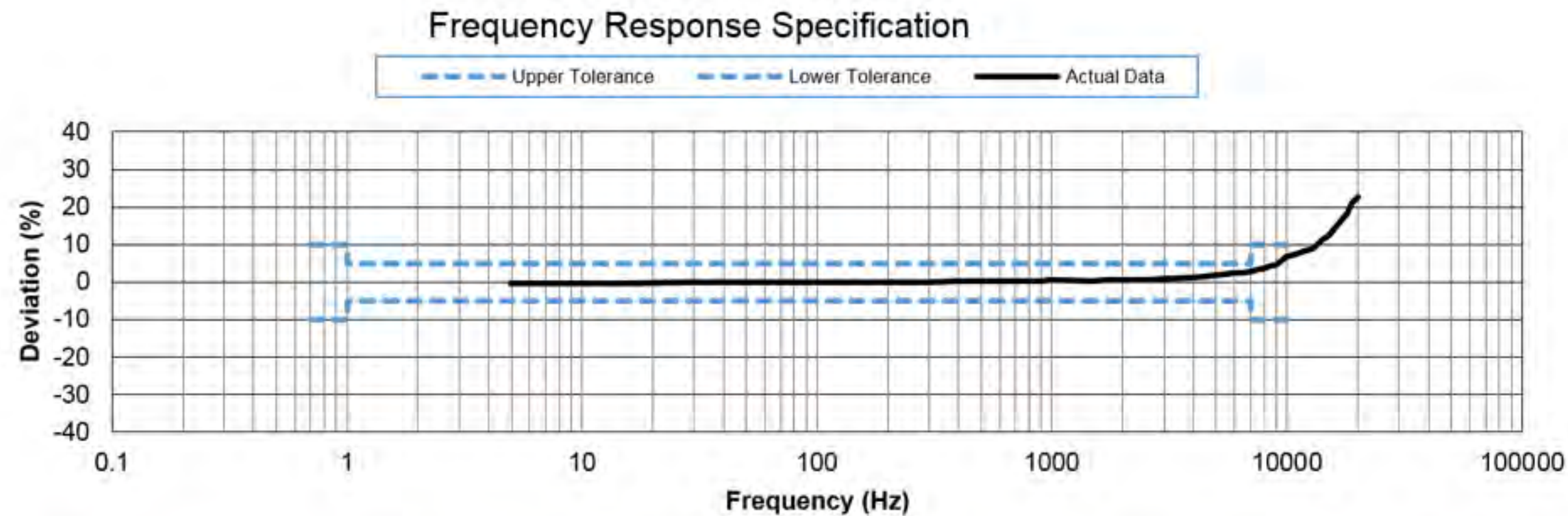
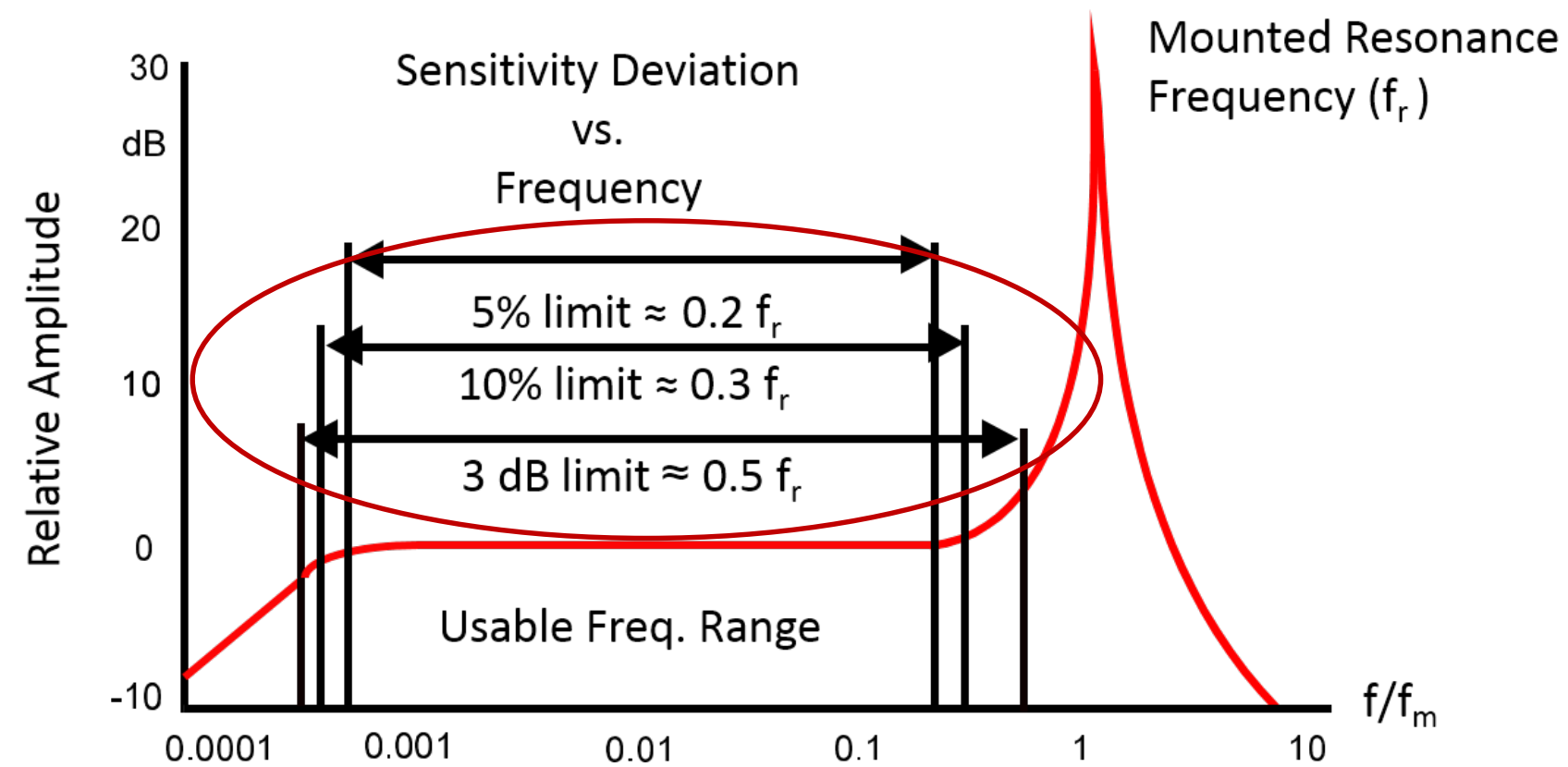
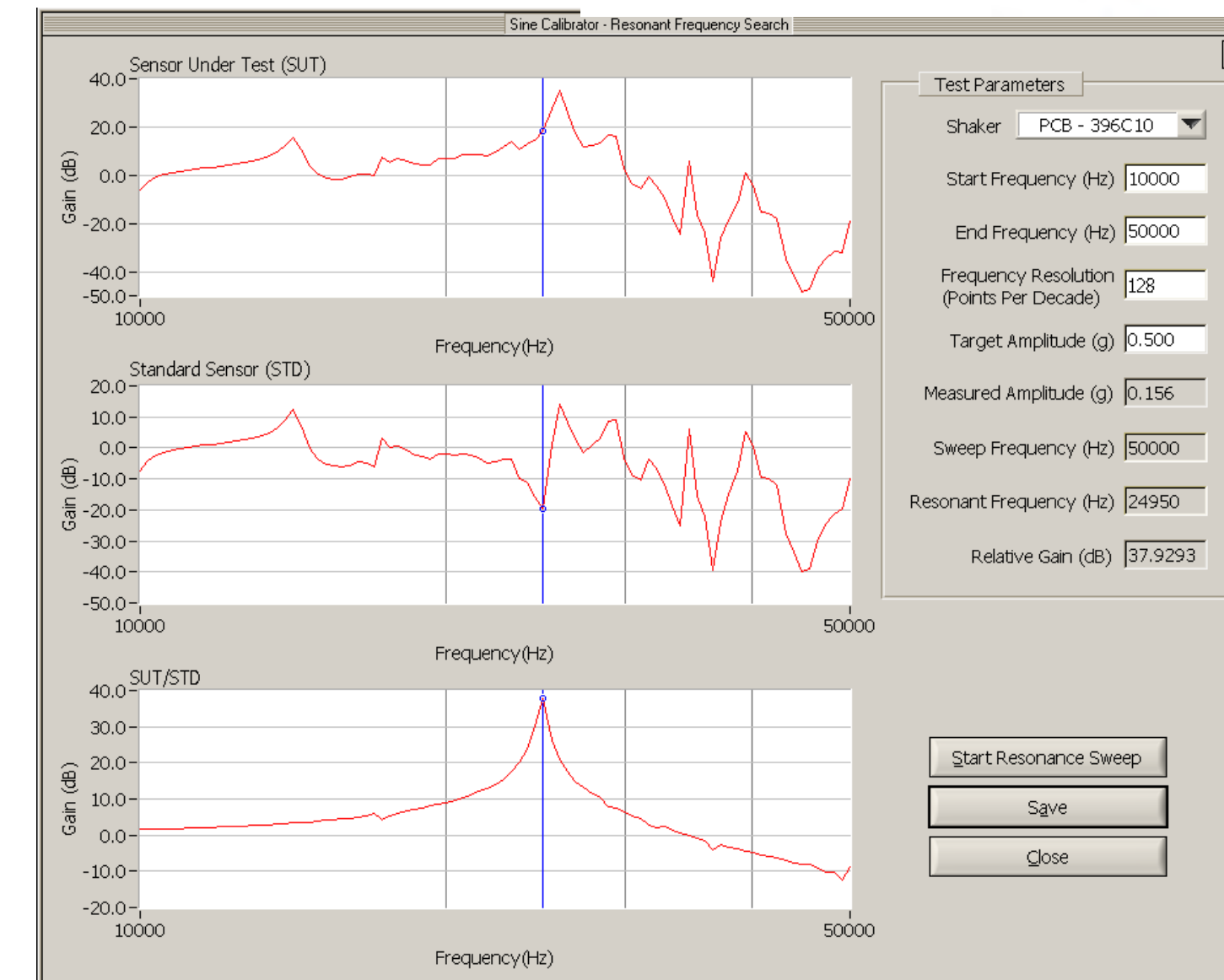
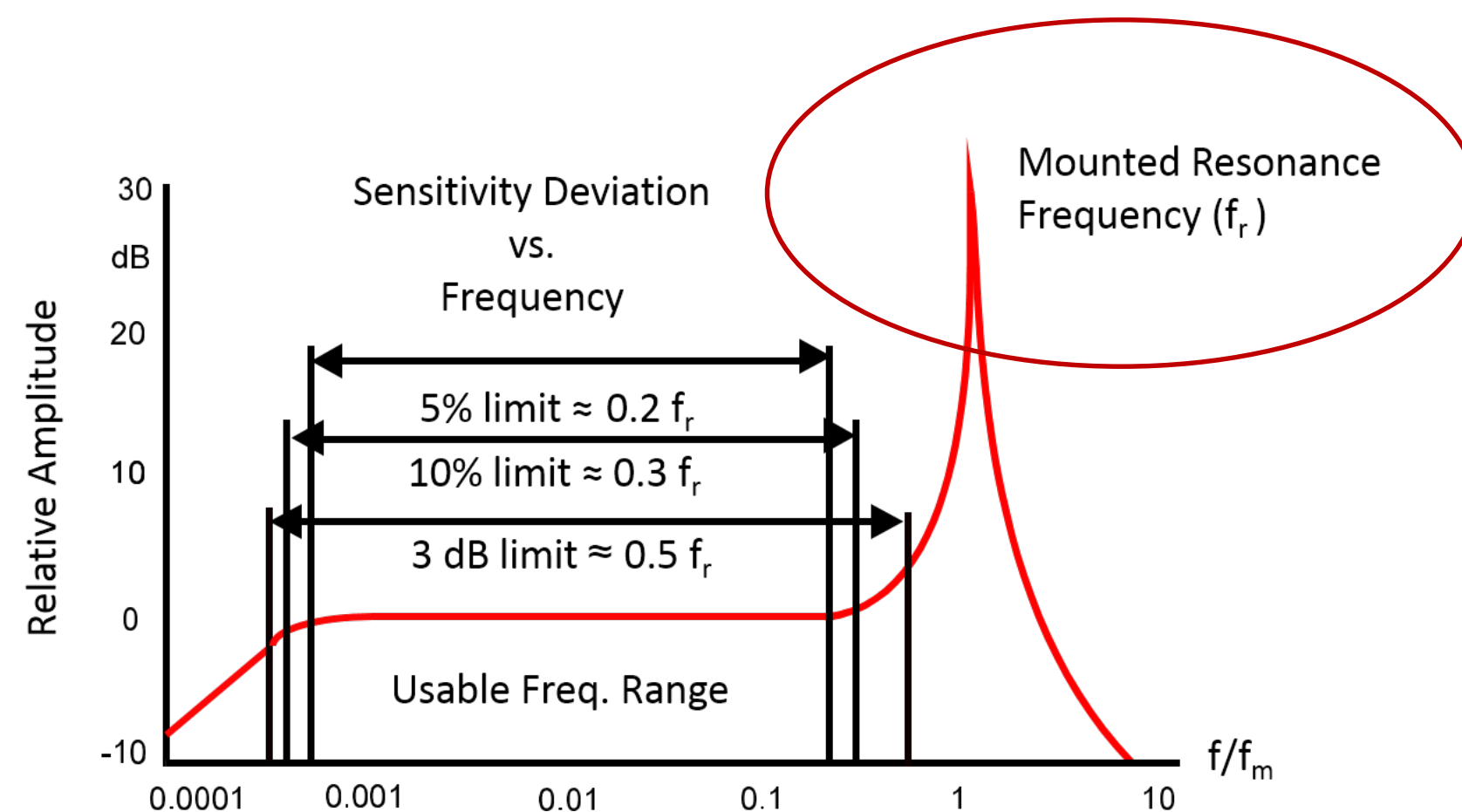


Figure 3 - Frequency Response Expressed Graphically. Tolerance Curves Correspond To Figure 2. These tolerances are applied to the 'Actual Data' as pass/fail criteria.

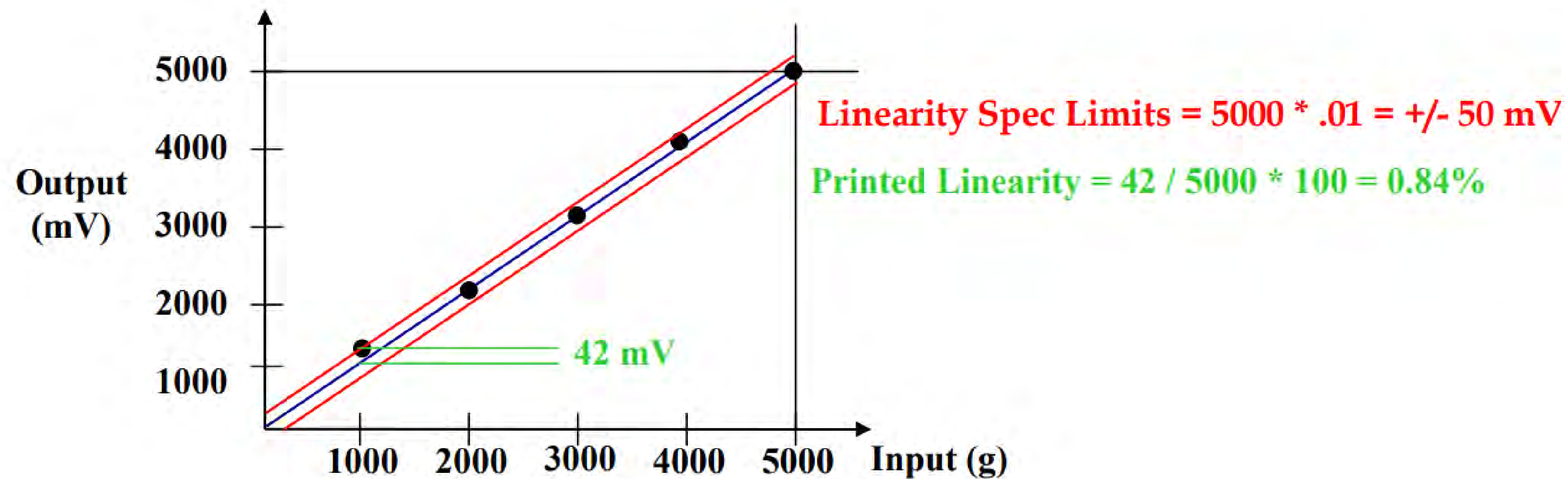
CALIBRATION METHODS: RESONANT FREQUENCY

- Stepped sine FRF measurement of test sensor to reference standard identifies the unit's mounted resonant frequency
- Verifying consistency of sensor's mounted resonant frequency provides additional assurance of sensor health



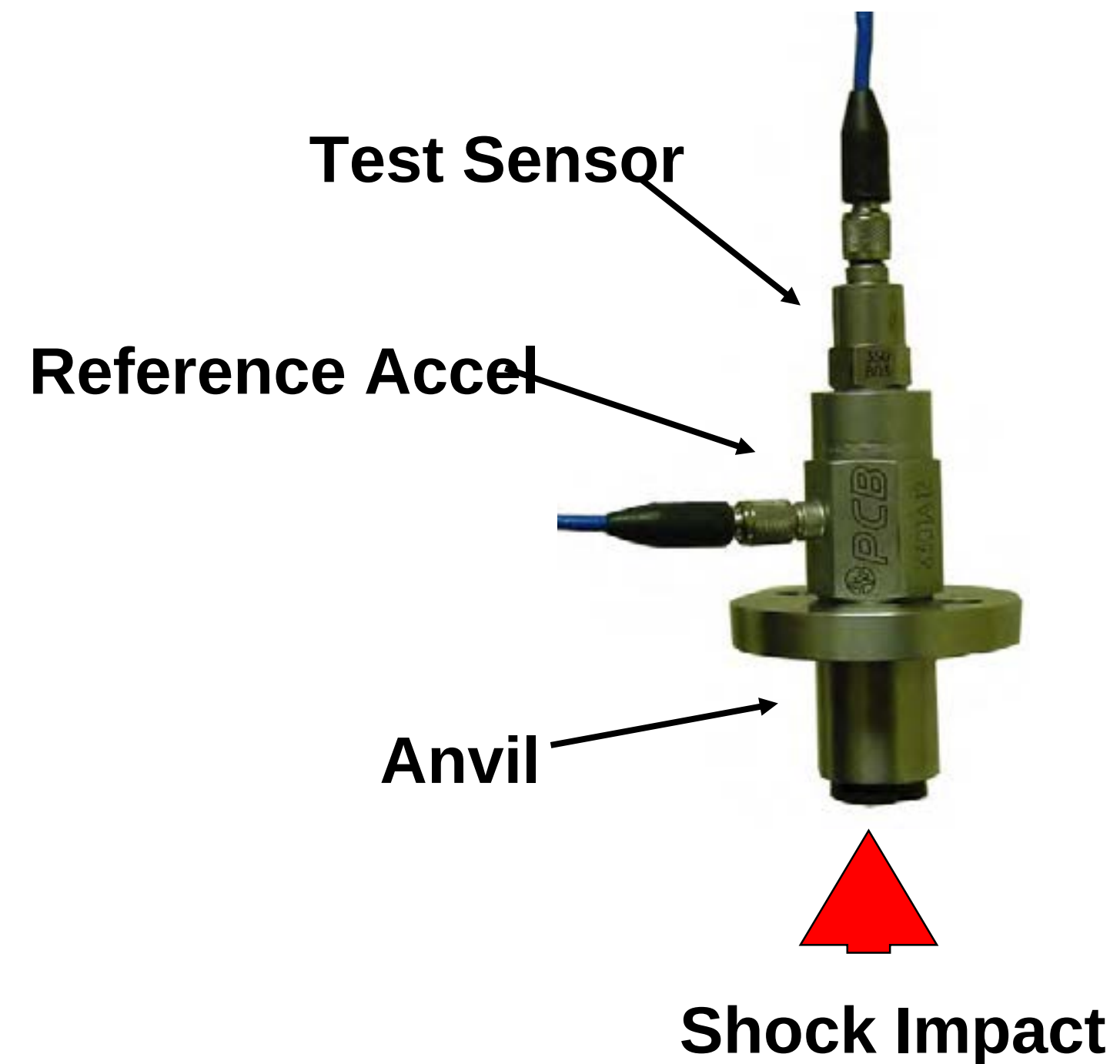
CALIBRATION METHODS: LINEARITY

- Change in sensitivity with change in test acceleration level
- Vibration shaker linearity in low amplitude regime



CALIBRATION METHODS: SHOCK LINEARITY

- Test sensor mounts “back-to-back” with a reference accelerometer and anvil, struck with a pneumatically operated projectile

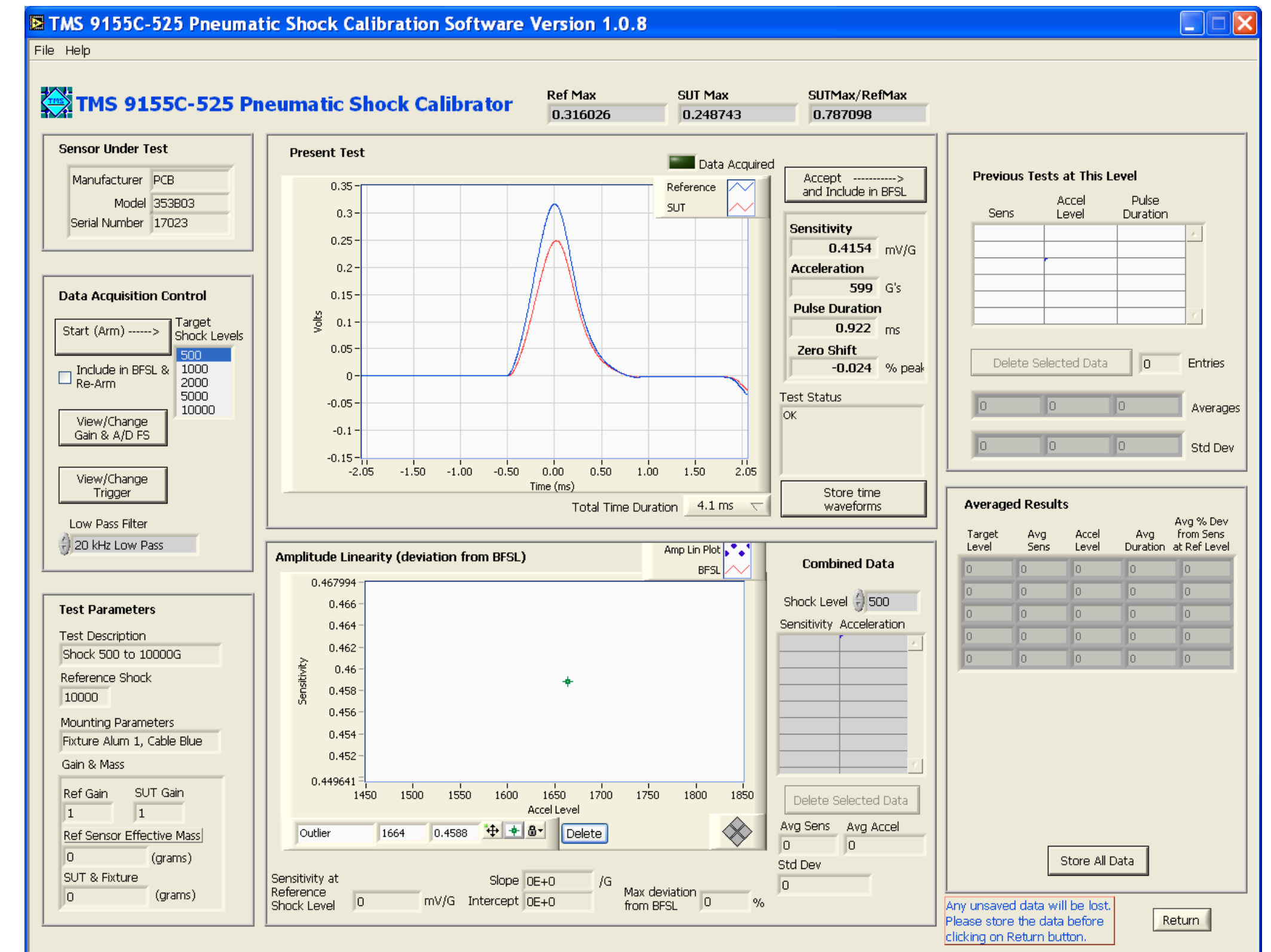


PneuShock™ shock tower

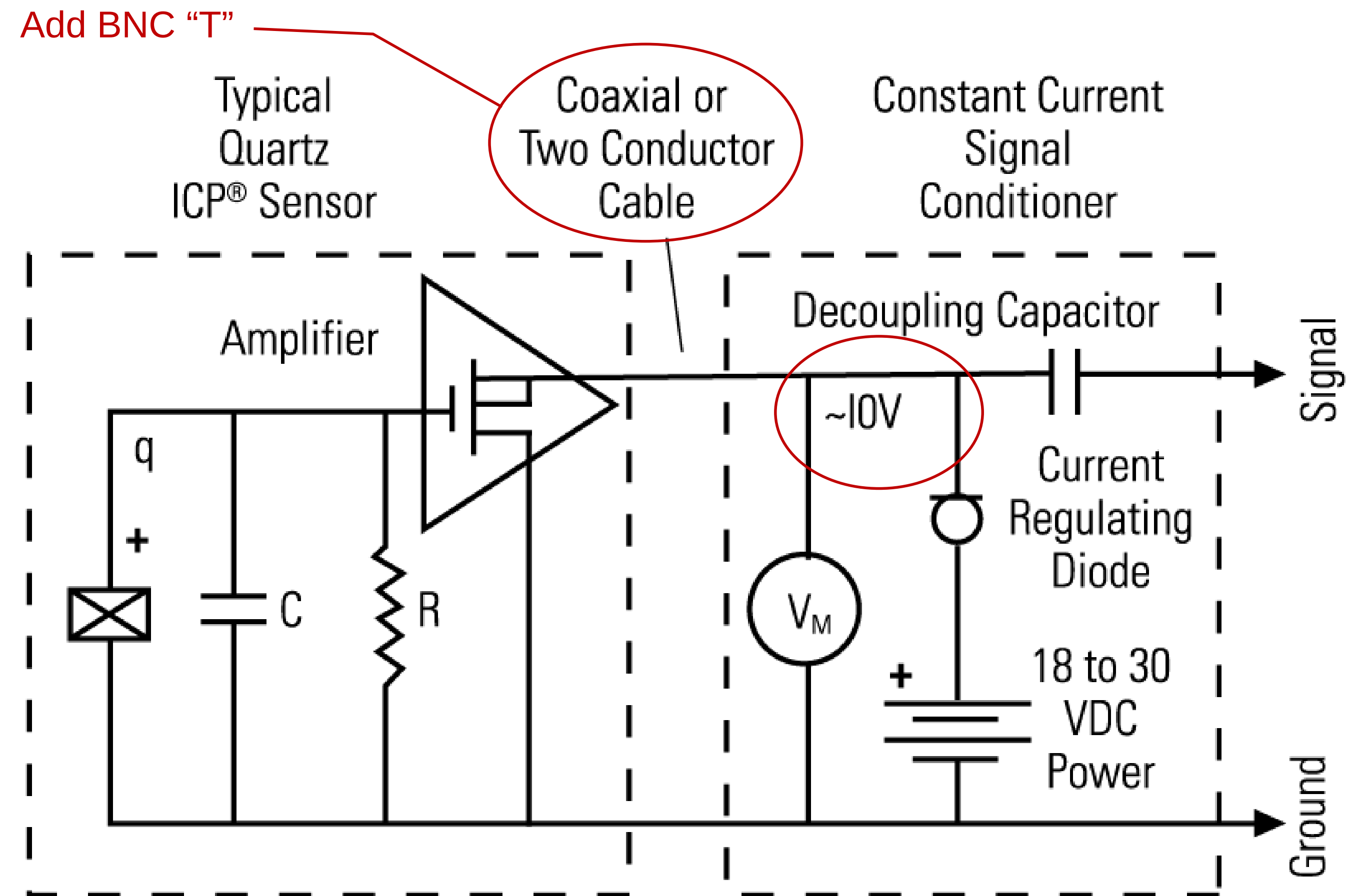


CALIBRATION METHODS: SHOCK LINEARITY

- ISO 16063 Part 22 shock calibration by comparison to a reference transducer
- Measures transient (impact) acceleration levels up to 10,000 g to calculate sensitivity
- Verifies amplitude linearity and performance of sensor at intended test levels

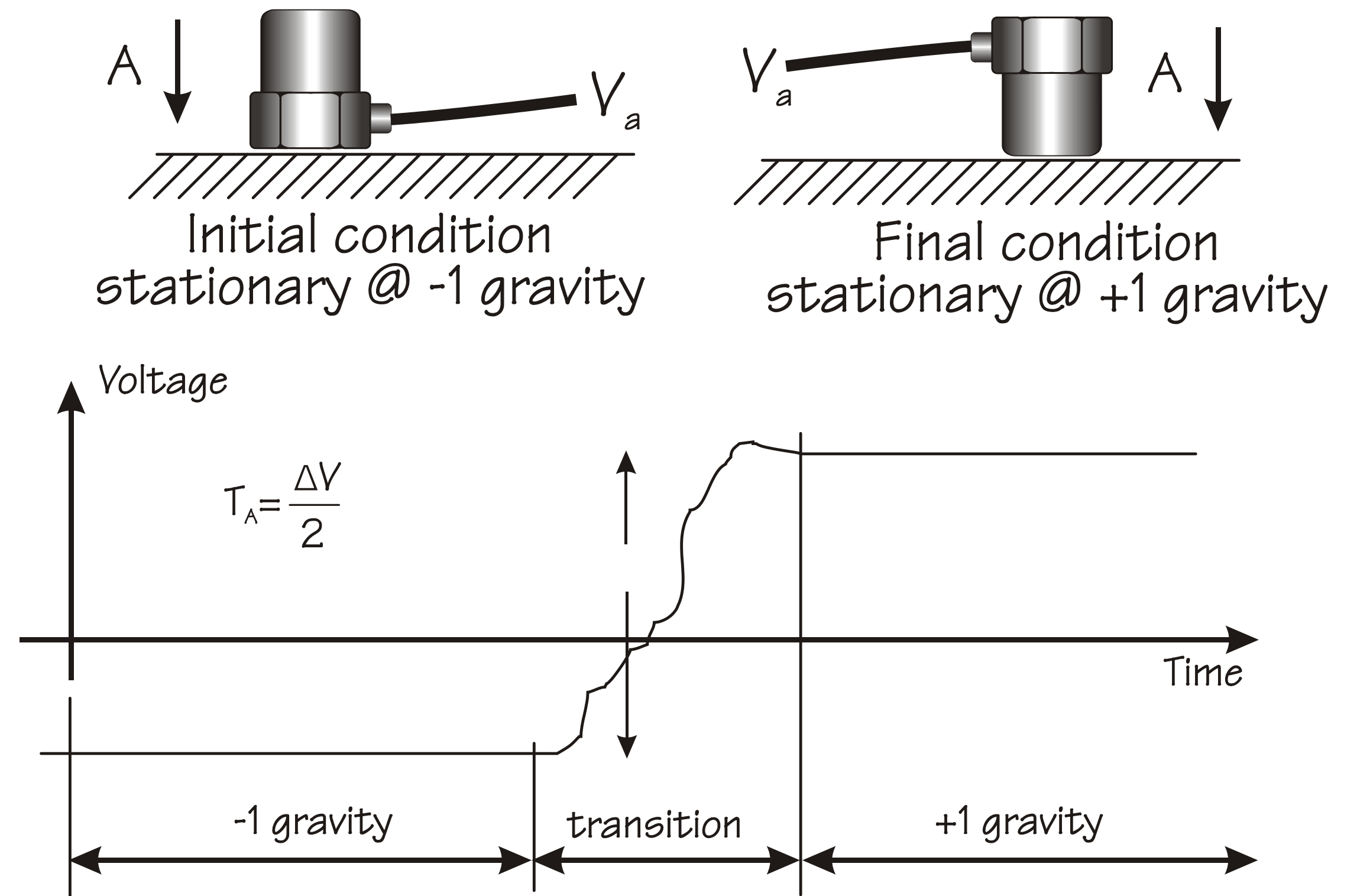


CALIBRATION METHODS: BIAS VOLTAGE



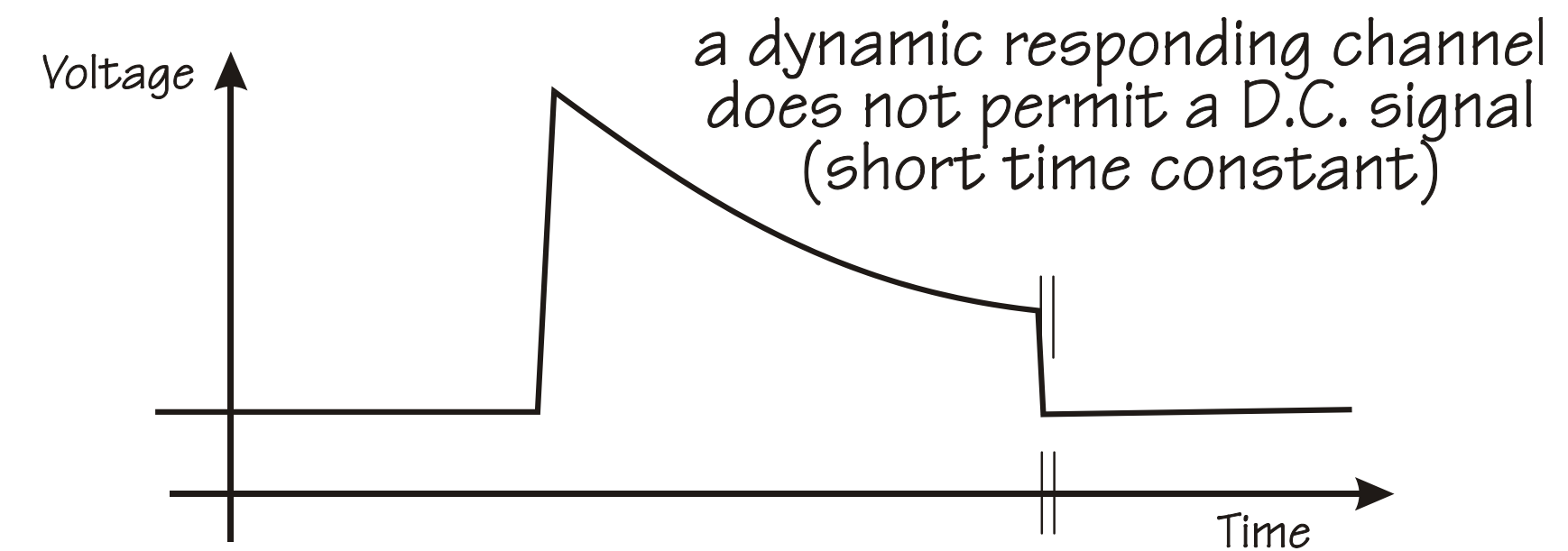
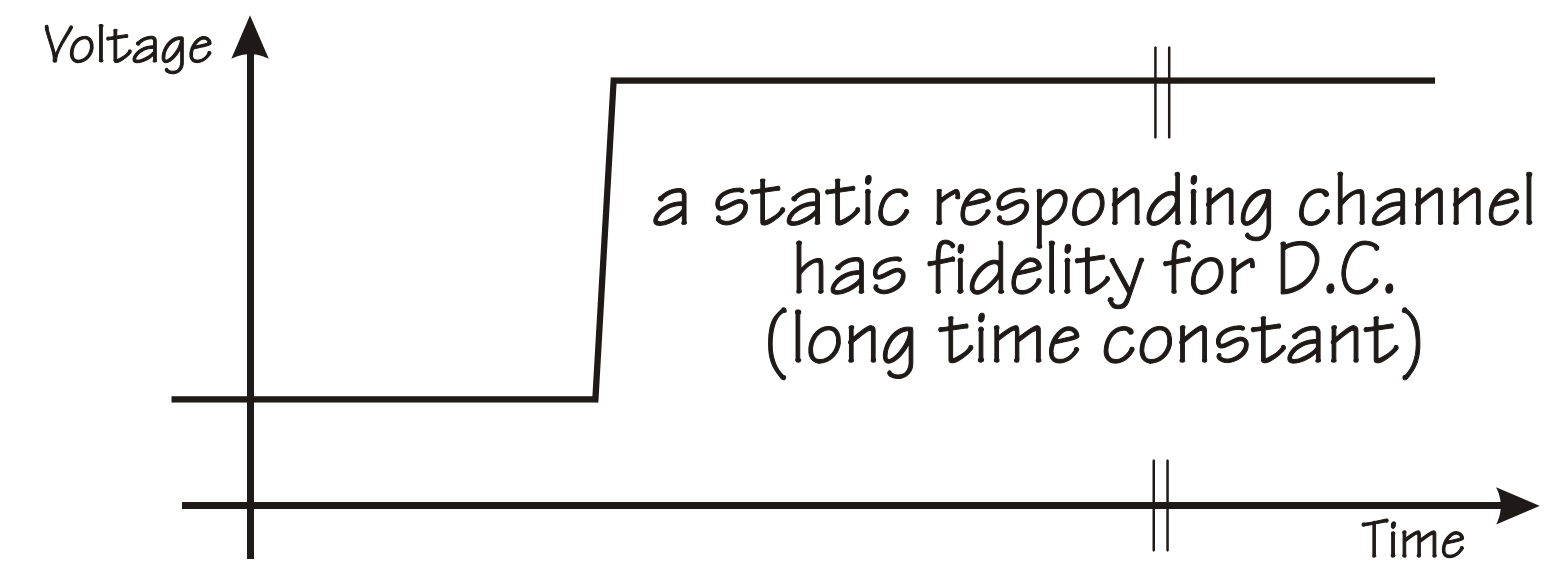
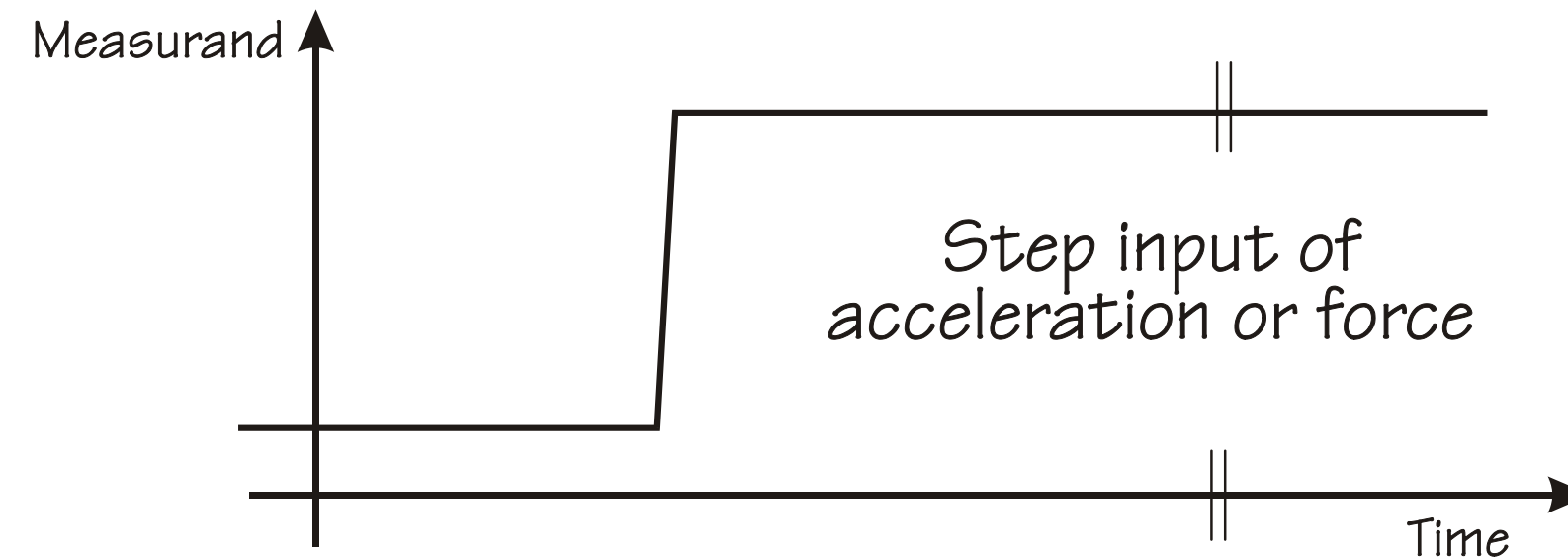
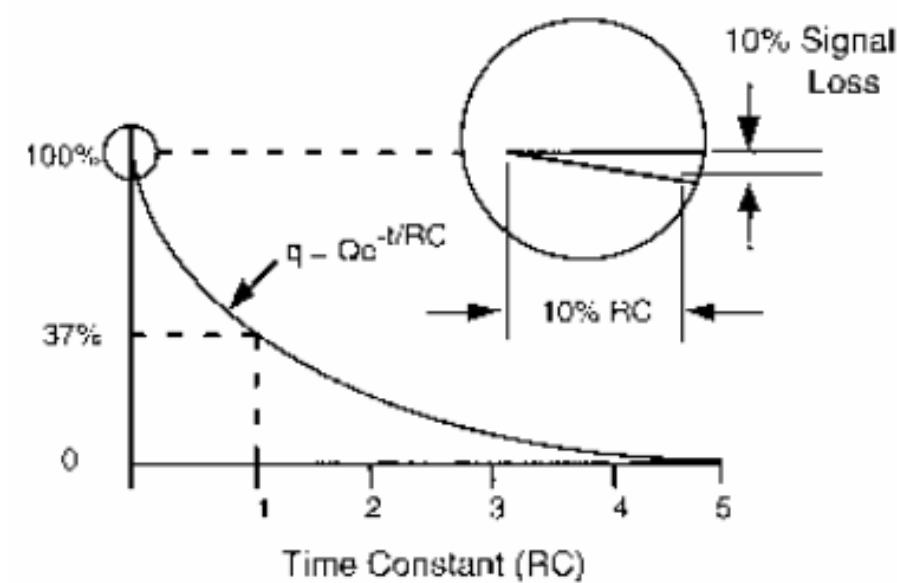
CALIBRATION: ZERO MEASURAND OUTPUT VOLTAGE

- Additional Horizontal position gives zero acceleration or “Zero Measurand” output.
- Indicator of measurement DC voltage offset error



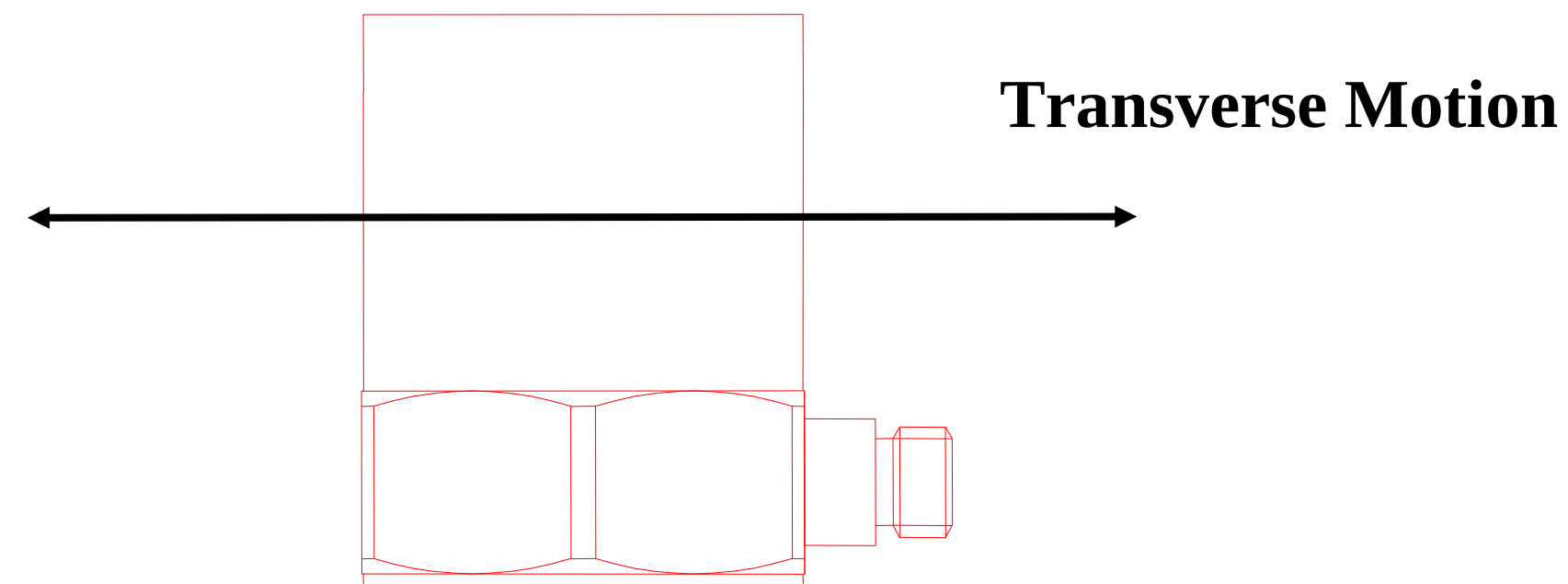
CALIBRATION METHODS: TIME CONSTANT

- Similar method to gravity inversion “flip” test, but faster
- Time to decay of ~63% of signal
- **Can be used to calculate the 3dB corner frequency response**
- DTC of signal conditioner can effect accelerometer response if shorter



CALIBRATION METHODS: TRANSVERSE SENSITIVITY

- Transverse sensitivity of accelerometer
- Sensitivity to motion perpendicular to the sensitive axis due to tolerance and alignment of sensing element

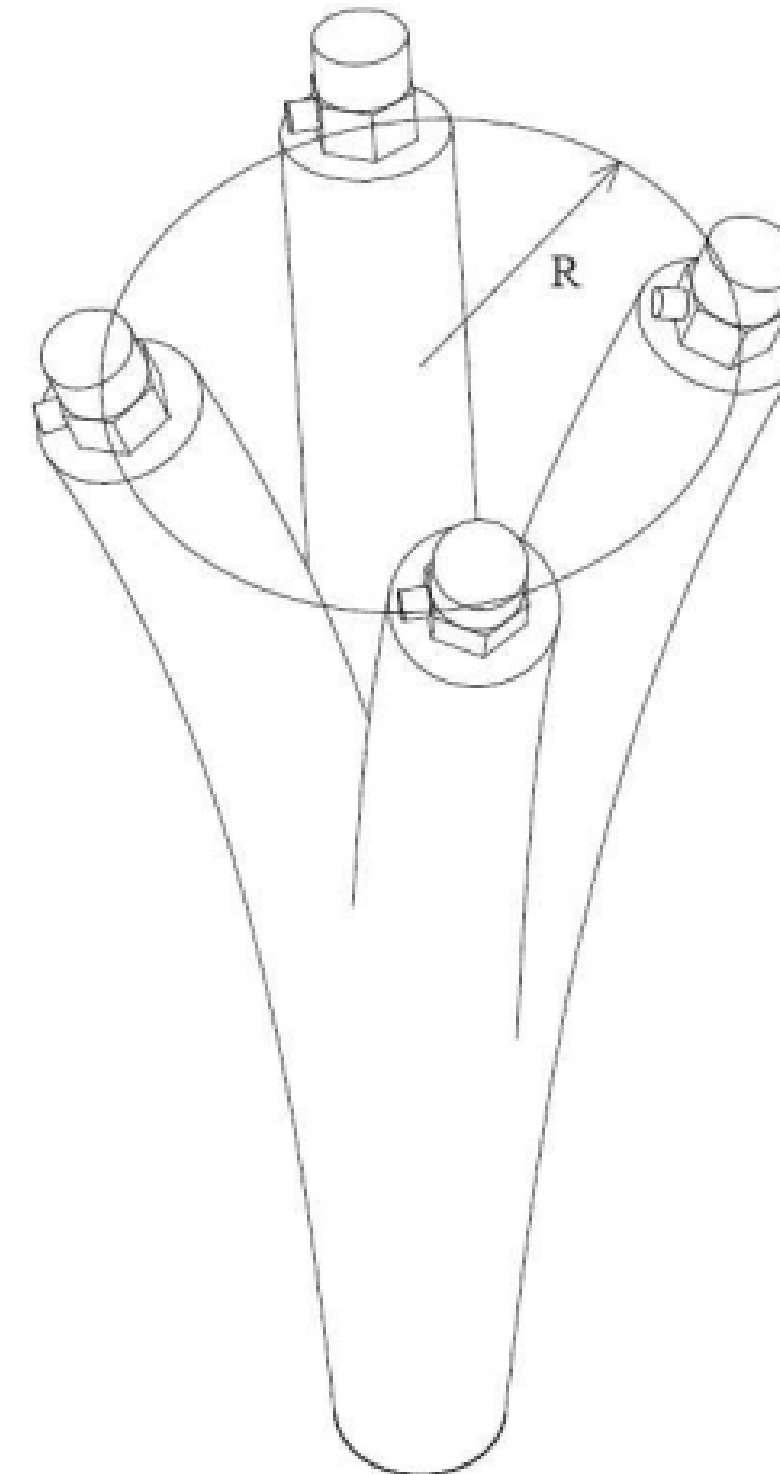
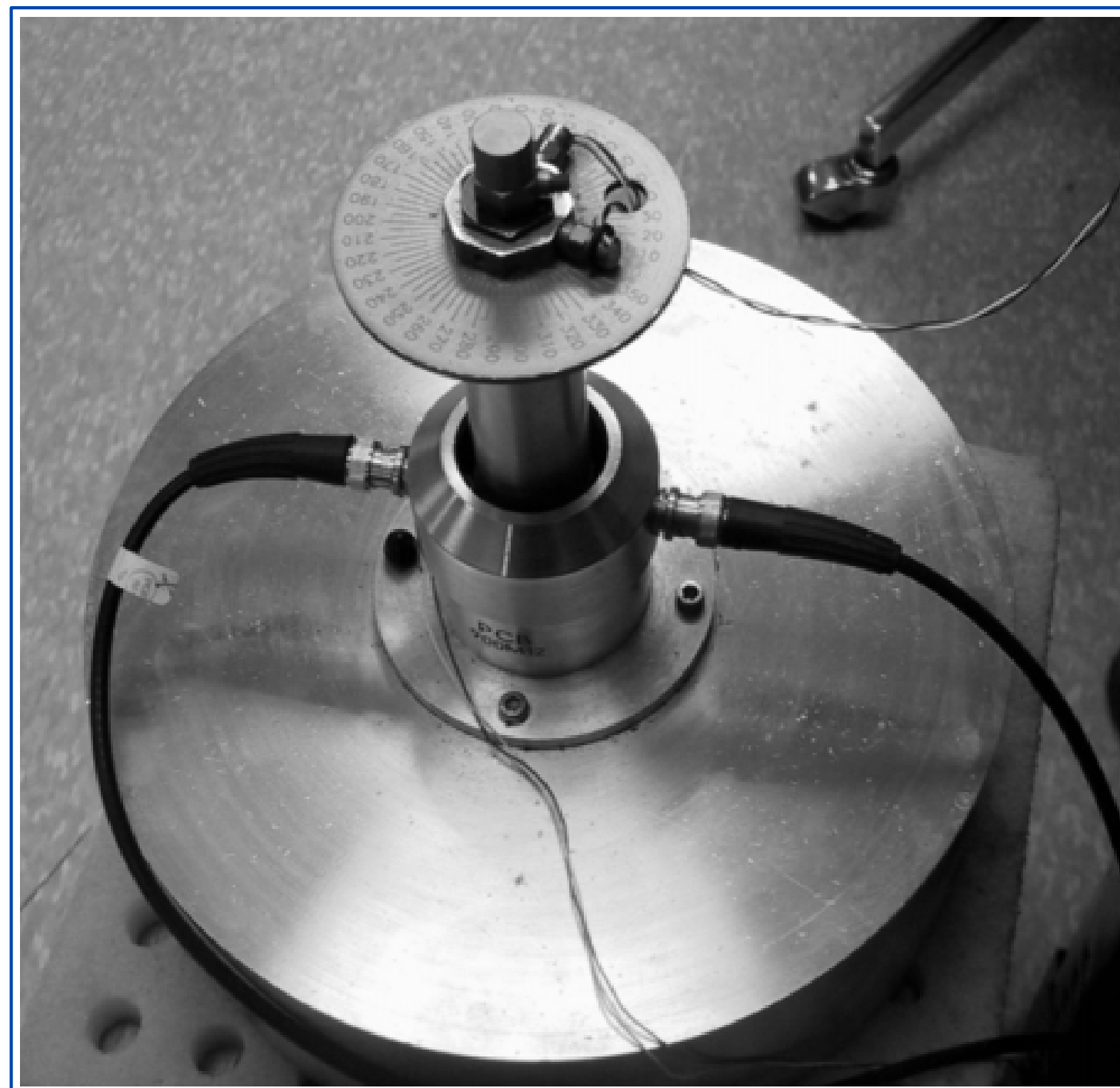


- Expressed as percentage of axial sensitivity

$$\% = \frac{\text{Transverse Sensitivity (mV/g)}}{\text{Axial Sensitivity (mV/g)}} \times 100$$

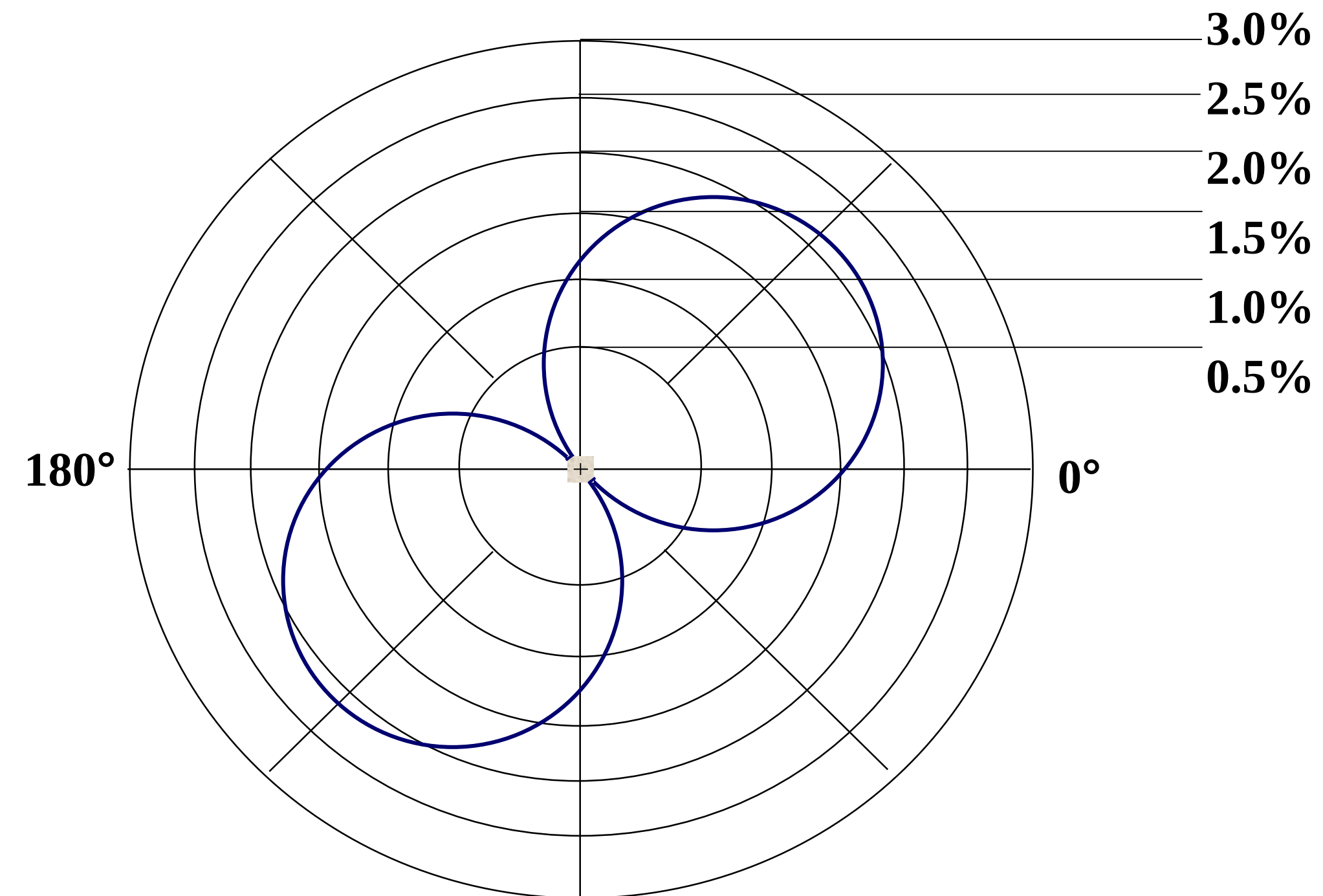
CALIBRATION METHODS: TRANSVERSE SENSITIVITY

- Acceleration Vector pointed towards center of circle
- Fixed test sensor mounting causes vector to rotate through 360 degrees relative to sensor



CALIBRATION METHODS: TRANSVERSE SENSITIVITY

- Transverse sensitivity presented as a polar plot showing sensor orientation corresponding to a max and min % value



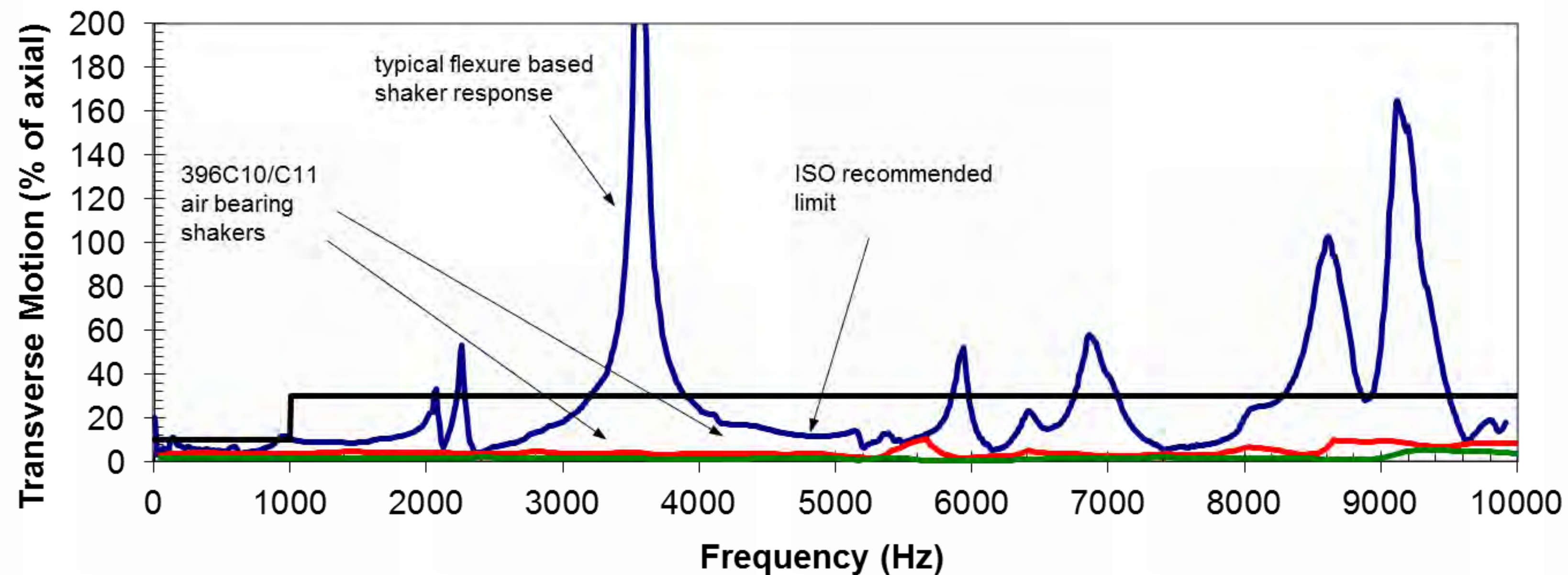
VIBRATION EXCITER

- Traditional flexure-based designs exhibit significant transverse motion
 - **Armature support system has lateral resonances well within typical frequency range for accelerometer calibration**
- Air bearing shaker meets ISO 16063 recommended limits for transverse motion
 - **Air bearing armature support system provides significantly more lateral stiffness**



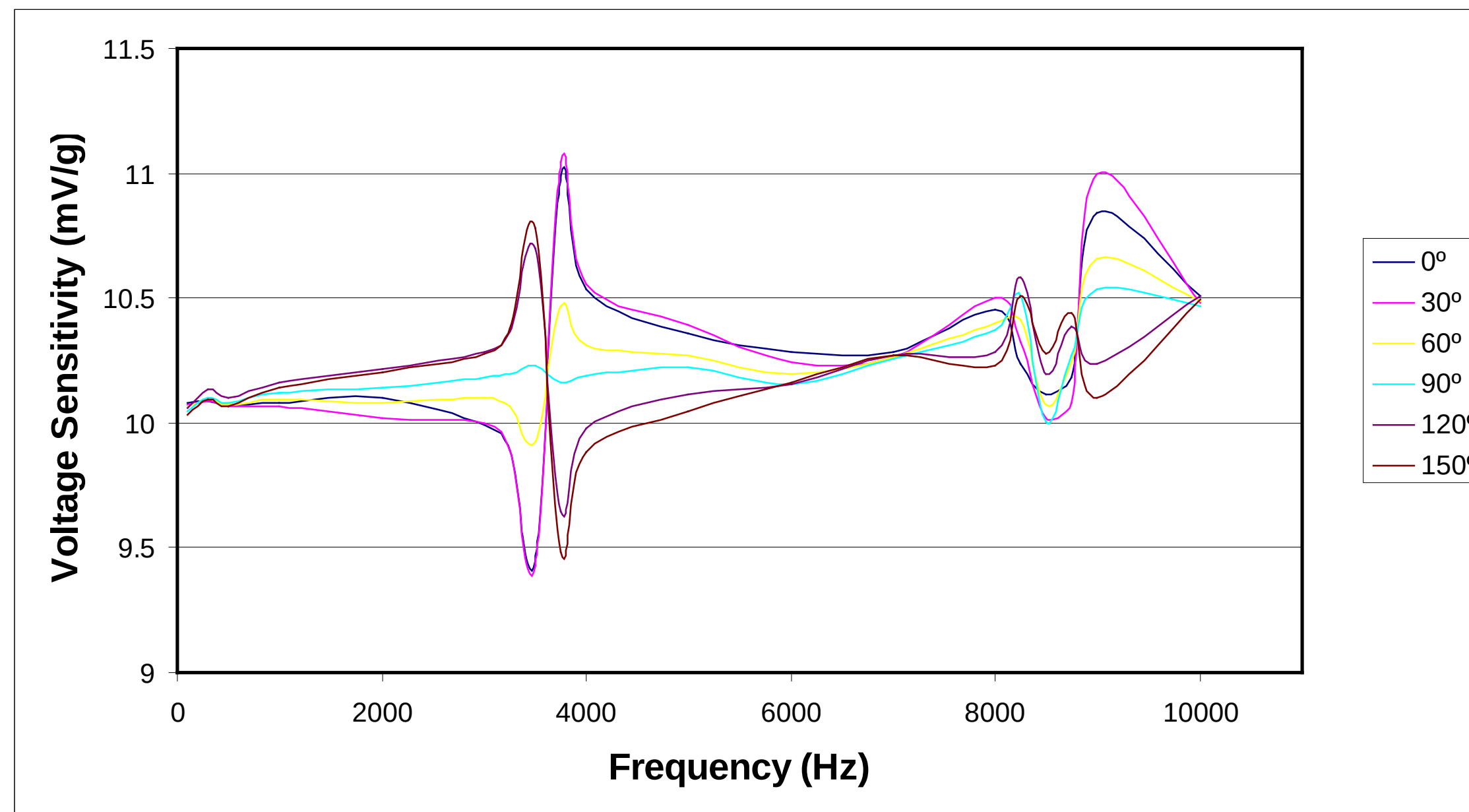
CONSIDERATIONS FOR EXCITER

- Transverse bending and rocking motion of shaker causes significant measurement error
- Calibration air-bearing shakers meeting ISO 16063 recommendations eliminate this error



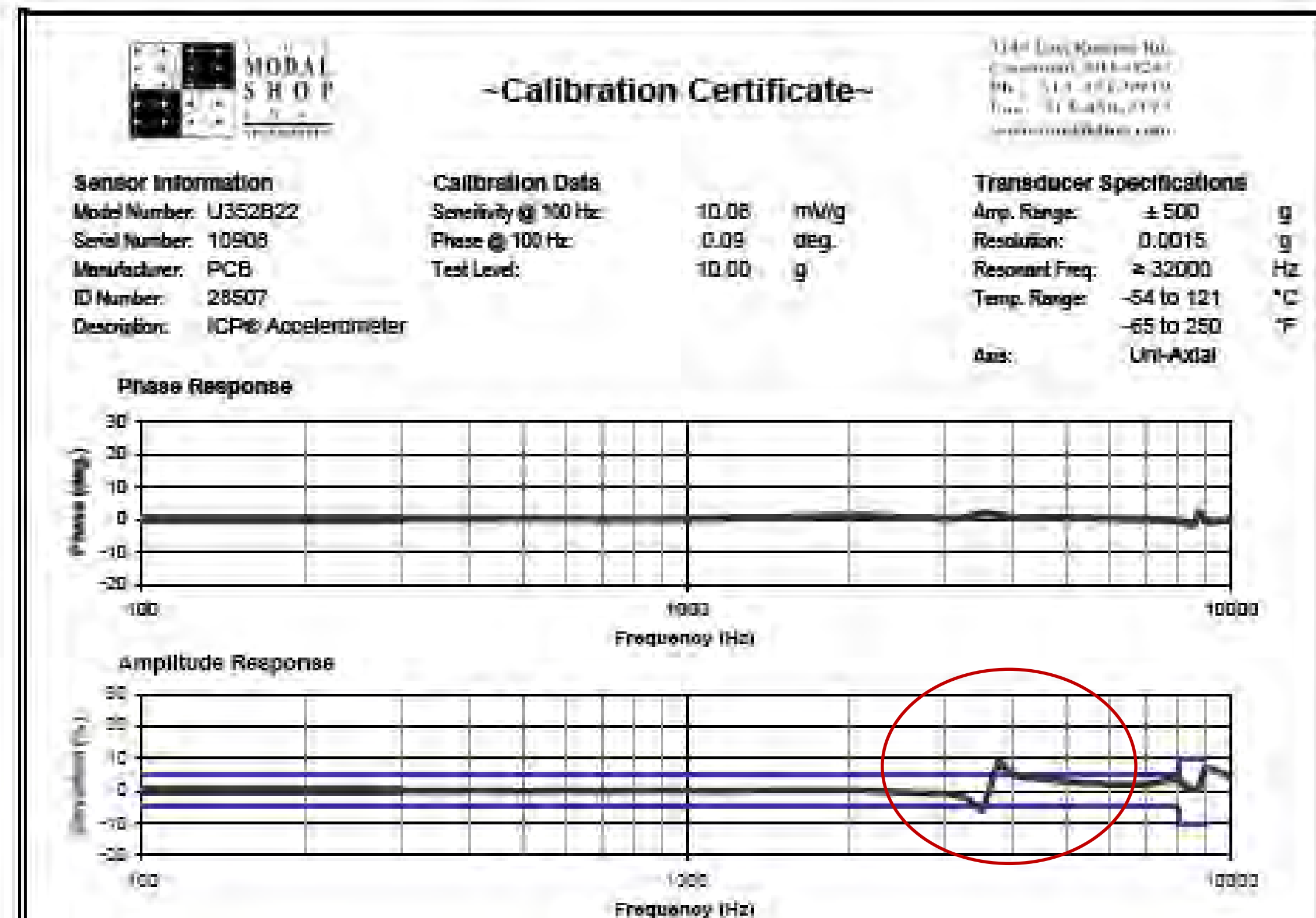
EFFECTS ON CALIBRATION RESULTS

- Data acquired using TMS 9155 calibration workstation with NI 4461 24 bit DAQ
- PCB 352B22 miniature teardrop ICP® accelerometer on flexure-based shaker by rotating it in 30° increments



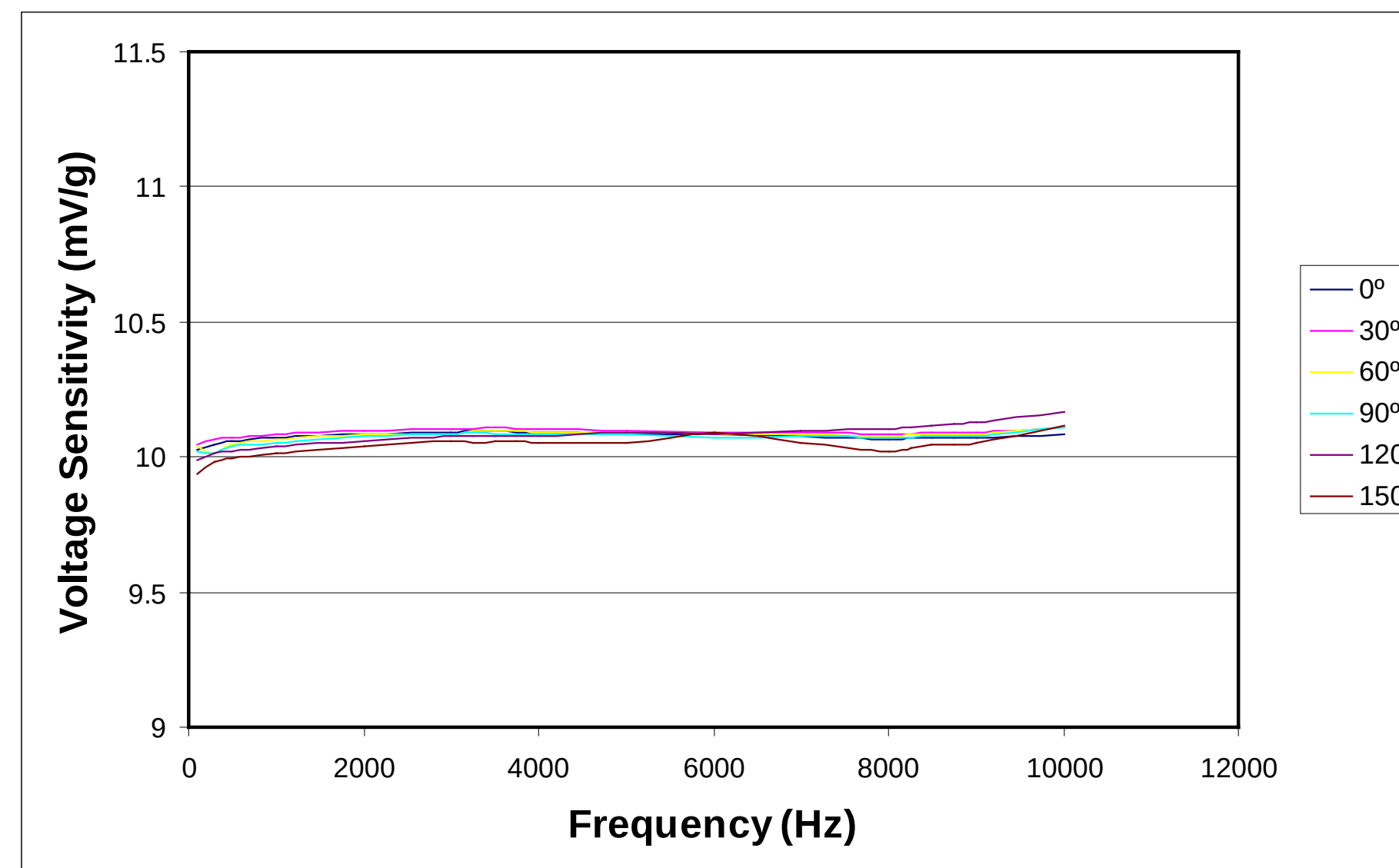
CONSIDERATIONS FOR EXCITER

- Uncertainty often misstated due to neglecting transverse motion effects, often as much as 3-5%



EFFECTS ON CALIBRATION RESULTS

- Data acquired using TMS 9155 calibration workstation with NI 4461 24 bit DAQ
- PCB 352B22 miniature teardrop ICP® accelerometer on air-bearing shaker by rotating it in 30° increments

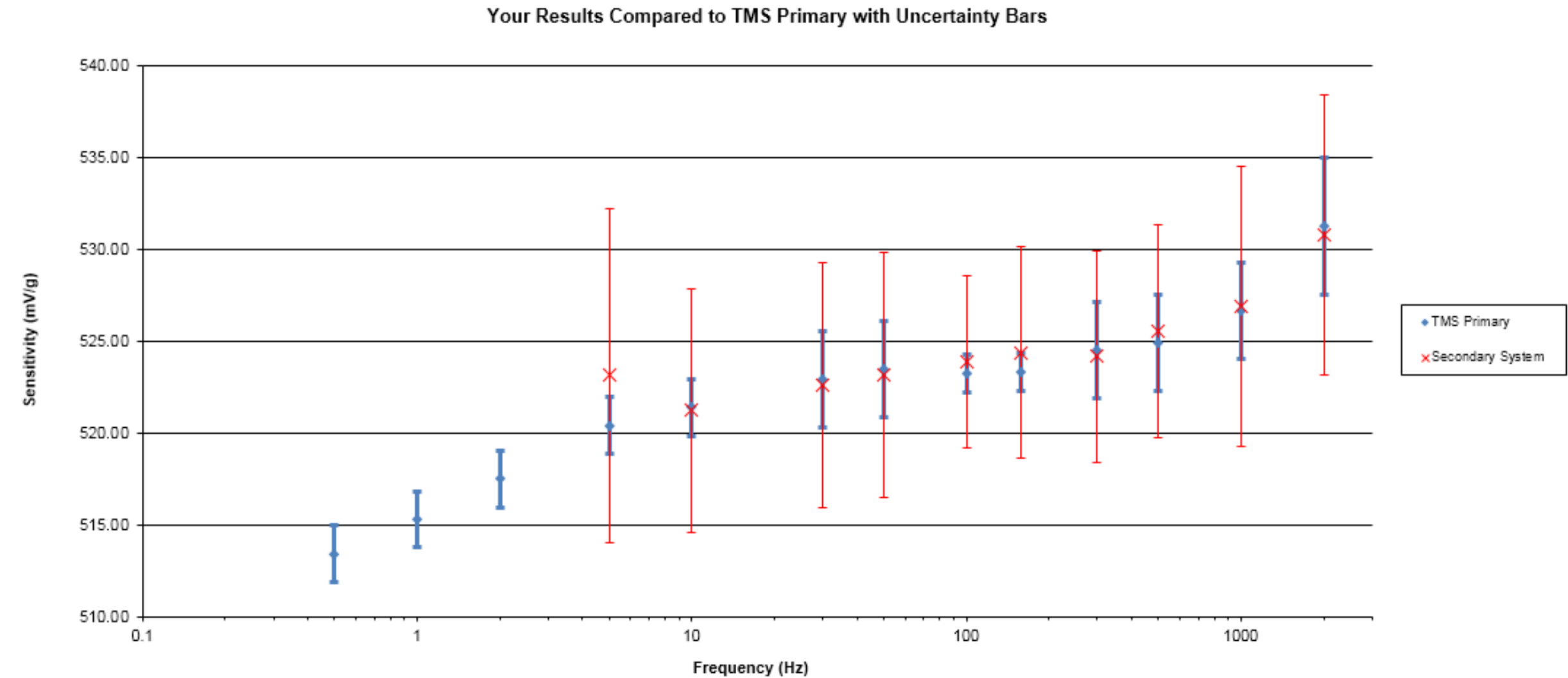


CALIBRATION OVERVIEW: STANDARDS

- What are the Standards?
 - ISO 9001 Company Quality Management System
 - ISO 17025 Calibration Laboratory Quality System
 - Identifies the general requirements for the competence of calibration laboratories
 - Accreditation bodies include A2LA, ACLASS, GCCRE, NVLAP, PCA, SCC, SWEDAC, etc.
 - ISO 16063 Methods for the Calibration of Vibration and Shock Transducers
 - 16063 Part 11. Primary vibration calibration by laser interferometry
 - 16063 Part 13. Primary shock calibration by laser interferometry
 - 16063 Part 21. Vibration calibration by comparison method
 - 16063 Part 22. Shock calibration by comparison method

MEASUREMENT UNCERTAINTY

- Calibration data is fairly meaningless without knowing the measurement uncertainty
- Systemic Component – Measurement Bias
- Random Component – Measurement Repeatability
- Ensure appropriate coverage factor



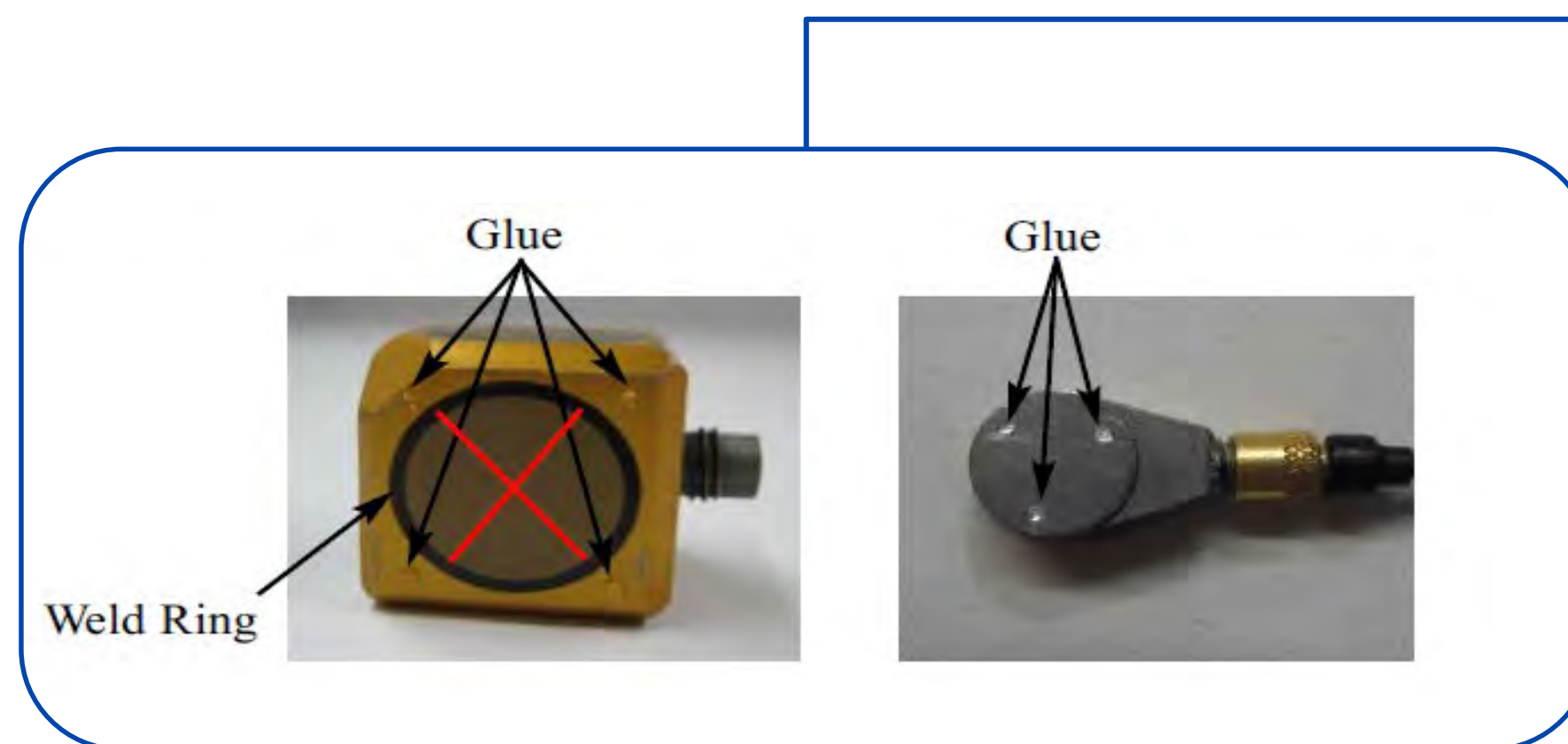
TYPES OF UNCERTAINTY

- **Systemic uncertainties**

- Reference and Signal Conditioner Uncertainty
- Transverse Motion of Exciter
- Reference Sensitivity “Drift”
- Measurement Channel Inaccuracies

- **Random uncertainties**

- Mounting and Cabling
 - **Torque Variation**
 - **Cable Strain Relief**
 - **Accelerometer Base Strain Sensitivity**
- Relative Motion Between SUT and Reference
- Operator Technique
- Electrical Noise
- Lab Conditions
 - **Temperature**
 - **Humidity**
 - **Pressure**



EXAMPLES OF UNCERTAINTY BUDGET

- Excerpt from The Modal Shop’s published ISO 17025 A2LA Certified Uncertainty Budget

Table 9: Uncertainty Budget

Uncertainty Component	Relative Uncertainty Contribution							
	5 to 9 Hz	10 Hz to 99 Hz	100 Hz	101 Hz to 920 Hz	921 Hz to 5000 Hz	5001 Hz to 10,000 Hz	10k Hz to 15k Hz	15k Hz to 20k Hz
Primary standard calibration	0.5	0.25	0.10	0.25	0.35	0.75	0.75	1.00
Primary standard stability	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06
Ratio Error	0.25	0.25	0.23	0.23	0.23	0.23	0.23	0.23
Shaker Transverse and Rocking Motion – 396C11	0.08	0.08	0.08	0.08	0.17	0.17	0.17	0.33
Random (Type A)	0.39	0.15	0.12	0.12	0.14	0.17	0.30	0.37
Combined Standard Uncertainty (%) $u_c - 396C11$	0.69	0.40	0.29	0.37	0.47	0.82	0.86	1.14
Expanded Uncertainty (%) $U (k=2) - 396C11$	1.38	0.80	0.59	0.74	0.95	1.64	1.71	2.28

Table 10: Published Best Expanded Uncertainty, $k = 2$.

Published Uncertainty (%) $U (k=2)$	1.65	1.15	0.75	1.00	1.35	1.85	2.20	2.75
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CALIBRATION: HANDHELD CHECK



- Handheld Calibrator
 - Quick, easy field check and/or calibration of entire system from sensor through cable to analysis equipment
 - Provides a fixed excitation and frequency
 - Battery operation provides for portable use, while DC adapters allows for extended operation
 - A handheld shaker is relative to an internal reference in shaker

CALIBRATION: FIELD CHECK



- Portable Vibration Calibrator
 - Quick, easy field check and/or calibration of entire system from sensor through cable to analysis equipment
 - Provides selectable amplitude and frequency settings for in-situ cal from 7 Hz to 10 kHz
 - Integral reference accelerometer provides traceability through comparison method
 - Supports acceleration, velocity and displacement

QUESTIONS?